TECHNICAL REPORT

Project Title:

Appointify:

A Data-Driven Analysis of Appointment Management in Service-Based Businesses

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2. Introduction

Purpose

This project presents an analytical study conducted as part of a capstone group project focused on the development of Appointify, a digital appointment and scheduling platform designed for small service-based businesses. The project integrates the efforts of multiple specialized teams, including data analytics, UI/UX design, frontend development, backend development, project management, and quality assurance, working collaboratively to deliver a functional, user-centered solution. The analysis establishes the foundation for understanding operational inefficiencies in appointment management and the role of data-driven insights in improving business performance and customer experience.

Objective of the Project

The objective of this analysis is to examine how appointment management practices, operational challenges, and levels of technology adoption influence customer turnout, customer satisfaction, and business willingness to adopt a centralized scheduling solution such as Appointify. The study aims to identify patterns, relationships, and behavioral trends within appointment-based businesses that can inform product design decisions, feature prioritization, and strategic planning for effective market adoption.

Problem Being Addressed

The problem addressed in this project arises from the widespread reliance on manual booking methods among small service-based businesses. Common practices such as WhatsApp messaging, phone calls, social media messages, and handwritten records often result in double bookings, missed appointments, delayed customer responses, inconsistent record-keeping, and the absence of automated reminders. These inefficiencies negatively impact customer experience, reduce operational efficiency, and limit business owners' ability to track performance, manage peak periods, monitor payments accurately, and plan for sustainable growth.

Key Datasets and Methodologies

The analysis is based on two primary datasets: the Business Plan dataset and the Customer Plan dataset. The Business Plan dataset captures operational attributes such as business type, booking methods, growth rate, booking challenges, technology adoption levels, administrative workload, and desired system features. The Customer Plan dataset focuses on customer-related outcomes, including turnout levels, satisfaction ratings, booking outcomes, urgency of need, and engagement patterns.

Microsoft Excel was used primarily for data cleaning, preprocessing, descriptive analysis, and the creation of PivotTables and summary dashboards. Power BI was used to develop interactive dashboards and advanced visualizations that highlight key trends, comparisons, and relationships across the datasets. This combined analytical approach supports accurate interpretation and effective communication of insights to both technical and non-technical stakeholders.

3. Story of Data

Purpose

The purpose of this analysis is to examine data generated from appointment-based service businesses and their customers in order to understand how current booking practices, operational structures, and technology usage influence customer experience, appointment outcomes, and adoption readiness for a centralized scheduling solution such as Appointify

Data Source

The data used in this project was generated from a structured survey designed specifically for this capstone project. The survey captured responses from both business owners and customers involved in appointment-based services.

Data Collection Process

Data collection was carried out over a period slightly exceeding one week. The survey was initially distributed to approximately 150 participants, including family members, workplace colleagues, and members of a fitness community. To increase response volume and diversity, participants were encouraged to share the survey within their networks. This approach enabled the collection of responses from a broader group of individuals, including both business owners and customers.

Data Structure

The dataset was organized into two related components: the Business Plan dataset and the Customer Plan dataset. In the Business Plan dataset, each row represents a business entity, while columns capture operational attributes such as booking methods, growth rate, challenges, and technology adoption. In the Customer Plan dataset, each row represents a customer response, with columns capturing turnout levels, satisfaction ratings, booking outcomes, and willingness to adopt a centralized booking system.

Important Features and Their Significance

The Business Plan dataset includes key features such as business type, booking challenges, reminder system usage, staff scheduling issues, payment tracking difficulties, technology adoption level, and desired features. These variables describe how businesses currently

manage appointments and internal operations.
The Customer Plan dataset contains features such as customer turnout, customer ratings, booking outcomes, and willingness to use Appointify. These variables provide measurable indicators of customer experience and engagement, allowing analysis of how business practices affect customer behaviour.

Data Limitations or Biases

The dataset has certain limitations, including potential sampling bias due to reliance on personal and extended networks during data collection. Since the data is self-reported, responses may reflect personal perceptions rather than objective measurements, introducing subjectivity and minor inconsistencies. Variations in respondents' understanding of operational and technological terms may also affect data consistency. Despite these limitations, the dataset remains suitable for identifying patterns and generating insights within the scope of this project.

4. Data Splitting and Preprocessing

This section explains how the datasets were prepared, refined, and structured to ensure accuracy, consistency, and suitability for analysis. The preprocessing steps were applied to both the Business Plan and Customer Plan datasets to establish a reliable foundation for subsequent analysis.

Data Cleaning
The initial data cleaning process focused on improving data quality and consistency across both datasets. Duplicate entries were reviewed and removed where repeated responses were identified. Inconsistent text entries, such as variations in booking methods, business types, and challenge descriptions, were standardized to ensure uniform categorization. Data formatting was also corrected to maintain consistency across columns, particularly for categorical variables used in comparisons and aggregations.

Handling Missing Values
Missing values were examined to determine their impact on the analysis. Where gaps were minimal and non-critical, affected records were retained without distortion of results. In cases where missing entries occurred in key categorical fields, Excel filtering and logical checks were used to exclude incomplete responses from specific analyses rather than the entire dataset. This approach preserved data integrity while minimizing loss of useful information.

Data Transformations
Basic data transformations were carried out to enhance interpretability and analytical clarity. Categorical responses were grouped into meaningful classes such as growth rate levels, technology adoption levels, and booking outcome categories. New analytical groupings were

also created to support comparison, such as aggregating similar booking challenges and aligning customer ratings into consistent scales for easier evaluation across businesses.

Data Splitting

The datasets were structured around independent and dependent variables to support relational analysis. Independent variables were primarily drawn from the Business Plan dataset and included factors such as business type, booking method, operational challenges, reminder system usage, staff scheduling issues, and technology adoption level. Dependent variables were drawn from the Customer Plan dataset and included customer turnout, customer ratings, booking outcomes, and willingness to use Appointify. This separation enabled clear comparison and correlation analysis between business practices and customer outcomes.

Industry Context

The data belongs to the service industry, specifically appointment-based businesses such as salons, wellness centers, healthcare providers, fitness professionals, tutors, and consultants. This industry relies heavily on efficient scheduling, customer engagement, and time management, making it well-suited for analysis related to digital booking and scheduling solutions.

Stakeholders

The key stakeholders of this project include the **Capstone project team @Vephla University, Academic mentors at Vephla University**, and appointment-based business owners who represent the primary end users of the Appointify platform. Within the project team, the findings are directly relevant to the data analytics team, UI/UX designers, frontend and backend developers, and project managers, as the insights support informed decision-making across product design, system development, and implementation. The outcomes of the analysis also indirectly inform customers of service-based businesses by contributing to improved booking experiences and service delivery.

Value to the Industry

The analysis provides valuable insights into how operational challenges and booking practices affect customer turnout, satisfaction, and adoption of digital solutions. These insights can help decision-makers within the service industry improve scheduling efficiency, reduce missed appointments, optimize internal operations, and support business growth through informed adoption of centralized appointment management systems.

5. Pre-Analysis

The pre-analysis phase focused on examining early patterns and observable trends within the Business Plan and Customer Plan datasets before conducting deeper analytical exploration. This stage provided an initial understanding of how business operations, booking practices, and customer behavior interact within appointment-based service environments.

Key Trends Identified

Across the Business Analysis dashboard, a strong pattern emerged showing that service-based businesses such as beauty salons, spas and wellness centres, massage therapy studios, and medical clinics dominate appointment usage. Among these, beauty salons consistently appeared as the highest category in customer turnout, service demand, and willingness to adopt Appointify.

Manual booking methods remained prevalent, with walk-in bookings, phone calls, WhatsApp messages, and social media direct messages accounting for the majority of appointment handling. Automated systems were noticeably less common, particularly among small and medium-scale businesses.

The dashboards also revealed that slow response time, overbooking, double bookings, misplaced appointments, and high no-show rates were the most frequently occurring booking challenges. These challenges appeared repeatedly across both business and customer views, indicating systemic operational inefficiencies rather than isolated issues.

From the Customer Analysis dashboard, high urgency booking requests were dominant, suggesting that customers often seek quick confirmations and immediate scheduling clarity. Customer discomfort factors such as lack of privacy, overcrowding, noisy environments, and unclear pricing also appeared frequently, reinforcing the importance of structured appointment management.

Potential Correlations

Early relationships were observed between reminder systems and customer turnout. Businesses that relied solely on manual reminders or had no reminder system showed higher missed and rescheduled appointment rates. In contrast, the presence of structured reminders appeared to align with improved booking completion.

Another noticeable relationship emerged between payment tracking challenges and administrative compliance costs. Businesses experiencing payment disputes, cash-only systems, or unclear pricing structures also reported higher administrative burdens and operational strain.

Technology adoption level showed a visible relationship with willingness to use Appointify. Businesses already familiar with basic digital tools demonstrated higher interest levels compared to those operating entirely through manual processes.

Initial Insights

Before deeper validation, the data suggested that operational inefficiency is directly linked to customer dissatisfaction and booking failures. The dashboards also indicated that businesses struggling with peak hour management tend to experience lower customer turnout and higher booking conflicts. These early insights framed the direction for deeper analysis in the next phase.

6. In-Analysis

The in-analysis phase involved deeper examination of patterns, category comparisons, and outcome distributions using structured dashboards created in Microsoft Excel and Power BI. This phase focused on exploring relationships across variables and forming testable insights.

Unconfirmed Insights

Analysis of business type by willingness to use Appointify suggested that beauty salons, spas and wellness centers, and massage therapy studios show the highest readiness for adoption. This appears to be driven by higher booking volumes and more frequent scheduling challenges, although further validation is required to confirm causation.

Booking challenges such as slow response time and overbooking appeared to correlate with lower customer turnout and higher cancellation rates. However, these relationships require additional longitudinal data to confirm consistency over time.

Customer rating distributions indicated that businesses with structured feedback processes and clearer booking workflows tend to receive higher satisfaction ratings. While this pattern is consistent across dashboards, it remains an inferred relationship rather than a statistically confirmed outcome.

Recommendations Based on In-Analysis

The analysis suggests prioritizing automated reminders as a core feature, as no-reminder environments consistently align with higher missed appointment rates. Integrating payment tracking and pricing clarity features would likely reduce administrative burden and improve booking outcomes.

Businesses with poor peak hour management should be targeted with scheduling optimization tools, as these periods are strongly associated with booking conflicts and customer dissatisfaction. Additionally, feature prioritization should focus on customer history records, payment dashboards, and cancellation control, as these consistently ranked highest in desired features.

Analysis Techniques Used

Microsoft Excel was used extensively for PivotTables, categorical comparisons, frequency counts, and cross-tabulation of independent and dependent variables. Excel dashboards were created to summarize booking challenges, growth goals, reminder systems, and outcome rates. Power BI was used to build interactive dashboards that visualized customer urgency levels, technology adoption patterns, booking challenge frequency, service type demand, and customer ratings. These tools enabled side-by-side comparison of business behavior and customer outcomes across both datasets.

7. Post-Analysis and Insights

The post-analysis phase consolidated findings from both datasets and assessed how the observed outcomes aligned with initial expectations.

Key Findings

The analysis confirmed that manual booking methods dominate appointment management among small service-based businesses, contributing to frequent booking challenges and customer dissatisfaction. Automated reminder systems were consistently associated with better booking outcomes and reduced missed appointments.

Beauty salons and wellness-related services emerged as the strongest segments in terms of demand, customer turnout, and willingness to adopt Appointify. High urgency booking requests and high rescheduling rates highlighted the need for faster confirmation workflows and real-time scheduling visibility.

Customer dissatisfaction was most strongly linked to no reminders, unclear pricing, long waiting times, and poor communication. Businesses that exhibited higher technology adoption levels showed stronger interest in centralized scheduling solutions and experienced fewer operational bottlenecks.

Comparison with Initial Findings

Initial expectations suggested that booking challenges would negatively impact customer turnout and satisfaction. The final analysis supported this assumption, with clear patterns showing that businesses facing frequent operational issues also recorded poorer customer outcomes.

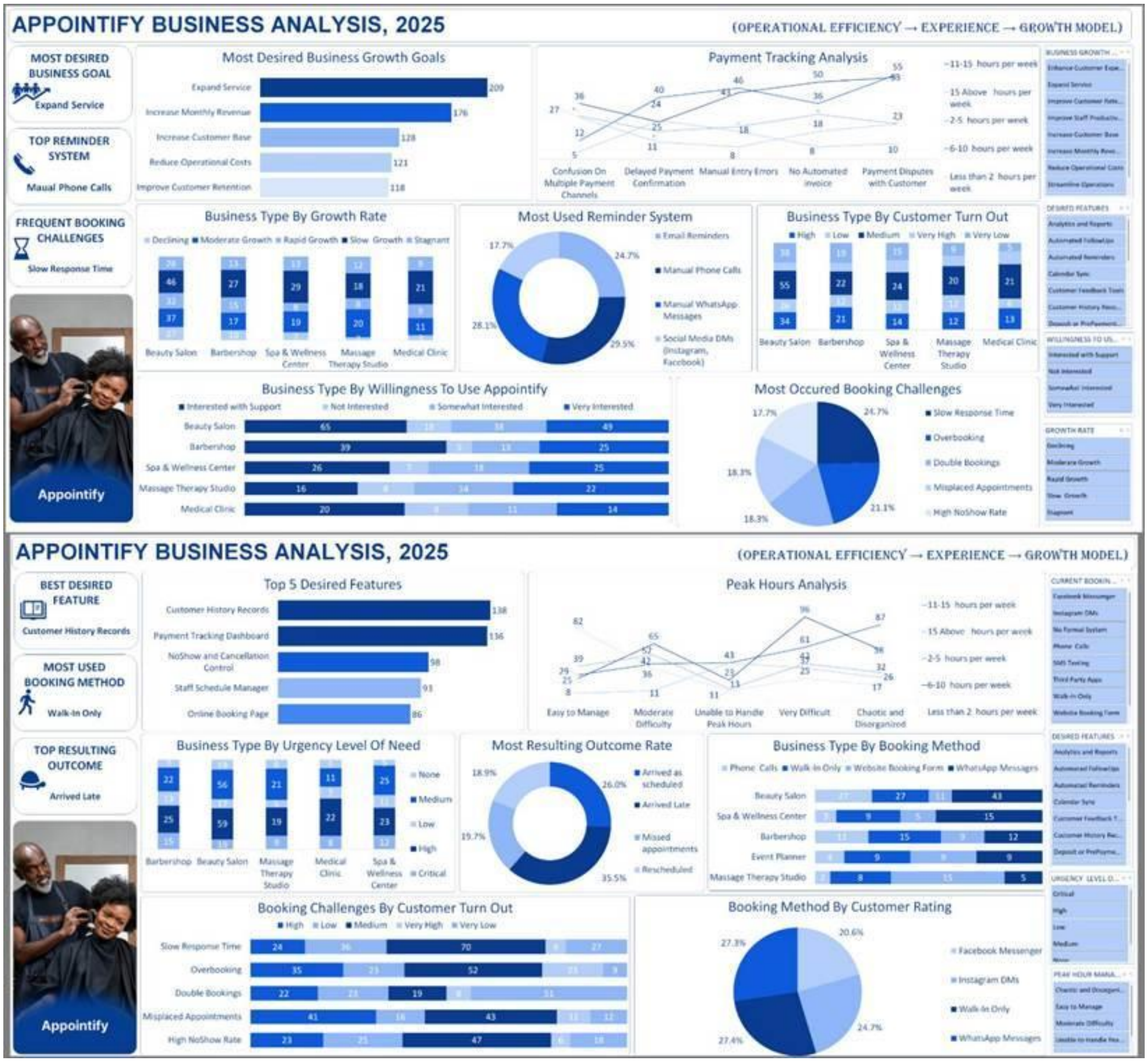
However, one notable insight was the high willingness to adopt Appointify even among businesses with low digital maturity, indicating that urgency of need may outweigh technology readiness in adoption decisions. This insight slightly contrasted with the initial assumption that only digitally mature businesses would show strong interest.

Overall, the findings validate the core premise of the project that a centralized, automated appointment management system like Appointify addresses critical operational and customer experience gaps within the service industry.

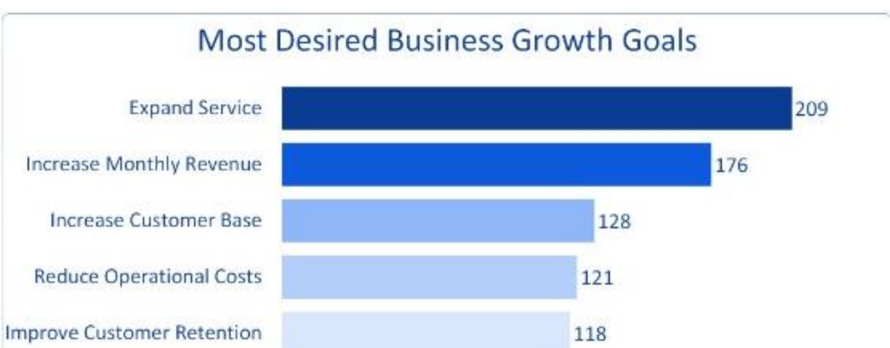
8. Data Visualizations and Charts

This section presents a comprehensive explanation of the visual analytics developed for the Appointify project. The visualizations were created using Microsoft Excel and Power BI to transform the Business Plan and Customer Plan datasets into interpretable insights. The dashboards consolidate key operational, behavioural, and adoption-related metrics, allowing stakeholders to quickly understand patterns, distributions, and relationships within the data.

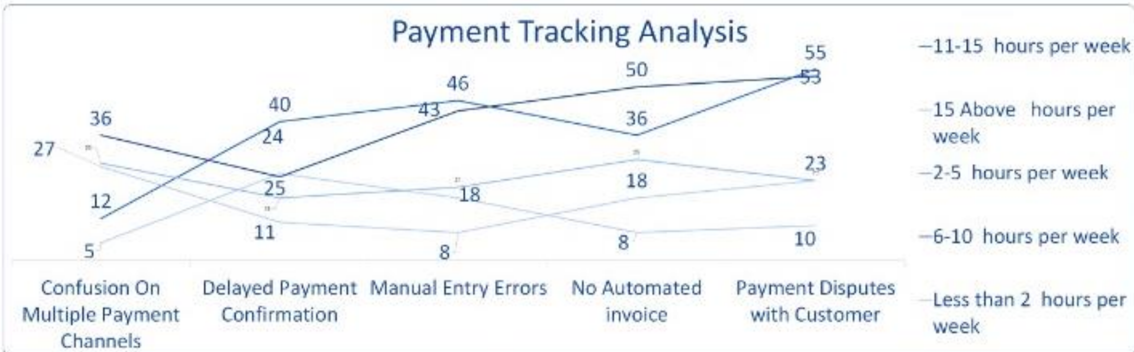
Appointify Business Analysis Dashboard 2025



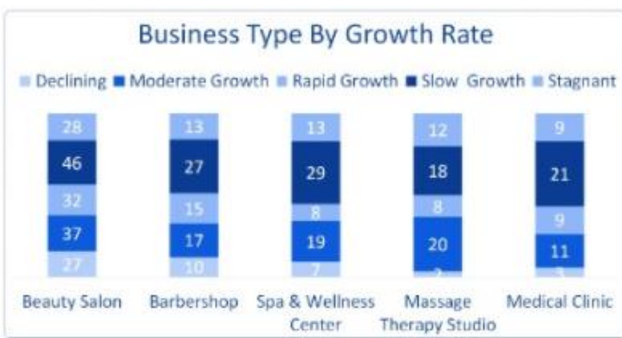
The Business Analysis dashboard focuses on operational efficiency, business growth objectives, booking processes, and adoption readiness among appointment-based businesses.



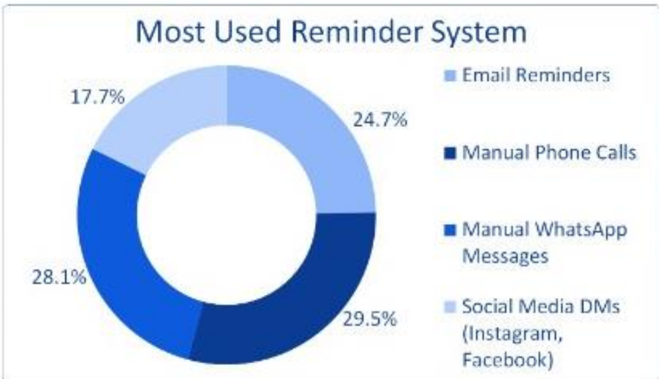
The chart titled Most Desired Business Growth Goals is a horizontal bar chart that ranks strategic priorities among businesses. Service expansion is the most selected goal with a count of 209, indicating a strong focus on scaling offerings. Increasing monthly revenue follows with 176 responses, while increasing customer base records 128. Reducing operational costs and improving customer retention record 121 and 118 respectively, suggesting that while efficiency is important, growth remains the primary driver.



The Payment Tracking Analysis is presented using a multi-line chart that compares payment-related challenges across varying weekly time commitments. Businesses spending more than 15 hours per week managing payments report the highest number of payment disputes at 55. Manual entry errors peak at approximately 50, while delayed payment confirmations exceed 40 across multiple time ranges. This visualization shows that increased manual effort does not reduce payment errors but instead reflects inefficiency in payment handling.

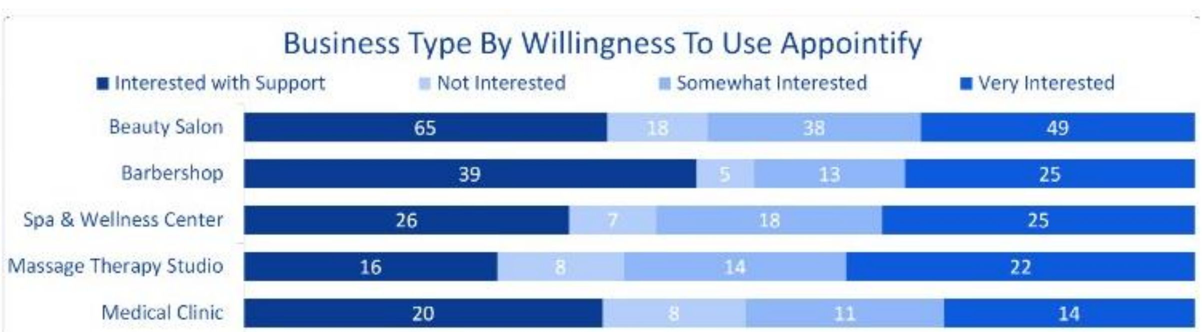


The Business Type by Growth Rate visualization is a stacked bar chart comparing growth categories across service types. Beauty salons show strong representation in moderate growth at 37 and rapid growth at 27. Spa and wellness centers follow similar patterns, while medical clinics show higher counts in stagnant growth at 21 and slow growth categories. This highlights variation in scalability across service types.



The Most Used Reminder System visualization is a donut chart showing reminder channel distribution. Manual WhatsApp reminders account for 29.5 percent, followed by manual phone calls at 28.1 percent. Email reminders account for 24.7 percent, while social media direct messages represent 17.7 percent. The chart clearly shows that over half of reminder communication remains manual.

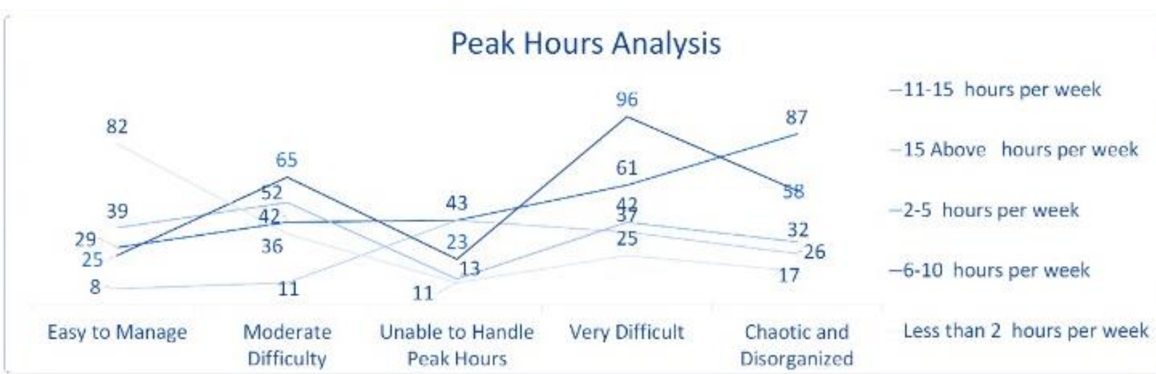
The Business Type by Customer Turnout chart is a grouped bar chart displaying turnout levels across service categories. Beauty salons record the highest medium turnout at 55 and high turnout at 38. Barbershops and spas show more moderate turnout distributions, while medical clinics demonstrate steadier but lower turnout variation.



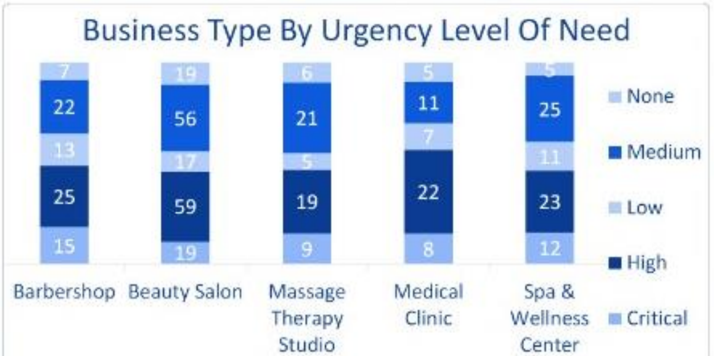
The Business Type by Willingness to Use Appointify visualization is a stacked bar chart categorizing interest levels. Beauty salons show the highest readiness, with 49 very interested and 65 interested with support. Spa and wellness centers and massage therapy studios follow closely, while medical clinics show more cautious adoption behaviour.

The Most Occurred Booking Challenges chart is a pie chart showing the distribution of operational issues. Slow response time is the leading challenge at 24.7 percent. Overbooking follows at 21.1 percent, while double bookings account for 18.3 percent. Misplaced appointments and high no-show rates form the remaining segments, confirming communication and scheduling inefficiencies.

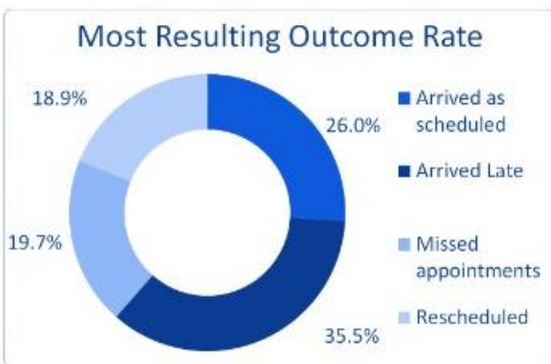
The Top 5 Desired Features visualization is a horizontal bar chart ranking requested functionalities. Customer history records lead with 138 requests, closely followed by payment tracking dashboards at 136. No-show and cancellation control records 98, staff schedule management 93, and online booking pages 86.



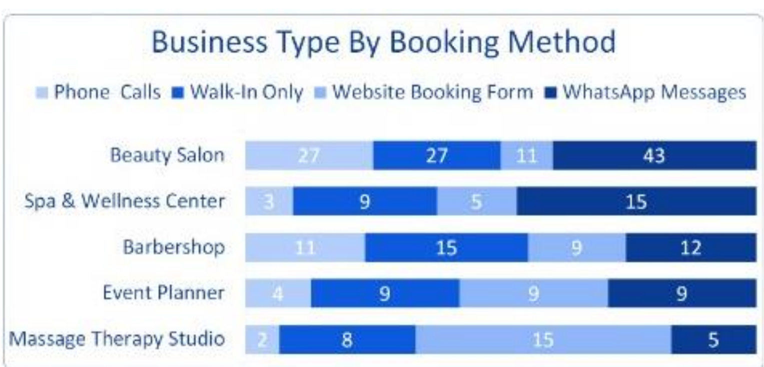
The Peak Hours Analysis is a multi-line chart examining difficulty levels across operational workloads. Very difficult peak periods record the highest difficulty count at 96, followed by chaotic conditions at 87. Businesses operating longer hours experience increased scheduling strain, particularly during peak demand windows.



The Business Type by Urgency Level of Need is shown using a stacked bar chart. Spa and wellness centers show the highest critical urgency at 25, while beauty salons record high urgency at 59. Medical clinics display more moderate urgency distributions.



The Most Resulting Outcome Rate is a donut chart summarizing booking outcomes. Only 26.0 percent of appointments occur as scheduled. Rescheduled appointments represent 35.5 percent, late arrivals 19.7 percent, and missed appointments 18.9 percent, highlighting workflow disruptions.



The Business Type by Booking Method visualization is a stacked horizontal bar chart. WhatsApp dominates across all service types, particularly in beauty salons with 43 bookings. Website booking forms and walk-ins follow, while phone calls show reduced usage.

The Booking Challenges by Customer Turnout chart is a stacked bar chart linking operational issues to attendance levels. Slow response time corresponds with the highest very high turnout loss at 70 cases. Overbooking and double bookings record 52 and 51 cases respectively, directly impacting customer attendance.

The Booking Method by Customer Rating visualization is a pie chart. WhatsApp bookings receive the highest ratings at 27.4 percent, closely followed by walk-ins at 27.3 percent. Instagram direct messages record 24.7 percent, while Facebook Messenger records 20.6 percent.

Customer Analysis Dashboard Visualization

Appointify Customer Analysis, 2025 (Excel Dashboard)

The Excel-based Customer Analysis dashboard provides a consolidated analytical view of customer behaviour, urgency patterns, comfort concerns, pricing perception, and readiness to adopt Appointify. The dashboard is designed to highlight customer-side pain points and adoption drivers, supporting decisions around scheduling automation, communication improvements, and experience optimization.

KPI Overview

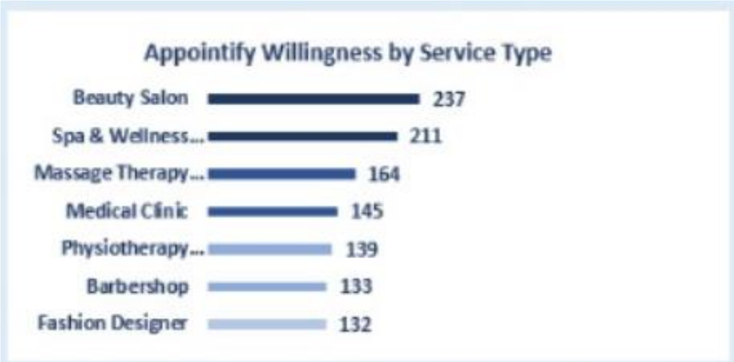
The KPI tiles at the top of the dashboard summarize the most critical customer indicators.

Top Booking Challenge is identified as **No Reminders**, with **114 responses**, indicating widespread absence of automated appointment confirmations.
Comfort Concern Leader is **No Privacy**, accounting for **40 percent**, highlighting environmental and service-setting discomfort.
Pricing Clarity Issue shows **Unclear Pricing** with **88 responses**, reflecting transparency gaps.
Top Service Type is **Beauty Salon**, with **237 users**, indicating the largest customer segment.
High Urgency Requests total **298**, confirming strong demand for time-sensitive bookings.
Total Willing Users number **1,161**, demonstrating high overall readiness to adopt a digital booking solution.

These KPIs establish a strong demand environment constrained by operational and communication weaknesses.

Appointify Willingness by Service Type

Chart Type: Horizontal Bar Chart



This chart displays the number of customers willing to use Appointify, segmented by service category.

Beauty salons lead with **237 willing users**, representing the strongest adoption segment.
Spa and wellness centers follow at **211**, while massage therapy studios record **164**.
Medical clinics and physiotherapy centers show moderate willingness at **145** and **139** respectively.
Fashion designers and barbershops record lower but still notable interest at **132** and **133**.

The distribution shows higher willingness in lifestyle and recurring-service industries.

Urgency Level Distribution

Chart Type: Vertical Bar Chart

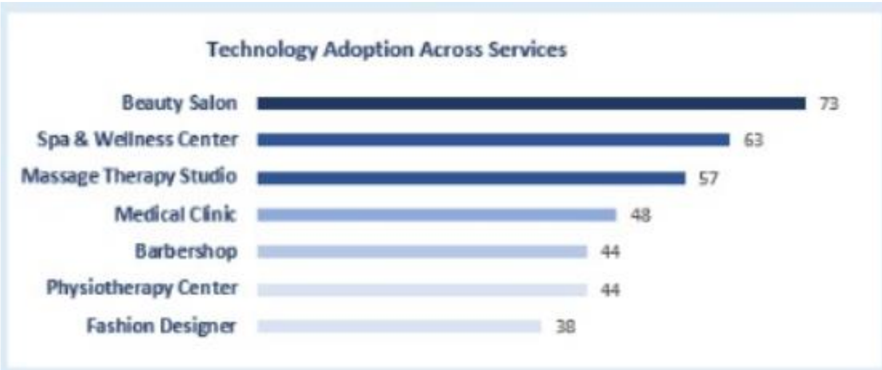
This chart shows how customer booking requests are distributed across urgency levels.

High urgency dominates with **298 requests**, indicating immediate booking needs.
Medium urgency follows at **244**, reflecting planned but time-sensitive appointments.
Critical urgency accounts for **222**, representing customers unable to tolerate delays.
Low urgency is minimal at **40**, confirming that most bookings are time-driven.

Urgency is a defining characteristic of customer expectations.

Technology Adoption Across Services

Chart Type: Horizontal Bar Chart



This chart compares technology adoption levels across service types.

Beauty salons record the highest adoption count at **73**.
Spa and wellness centers follow with **63**, and massage therapy studios record **57**.
Medical clinics and barbershops show moderate adoption, while fashion designers record the lowest at **38**.

Technology readiness aligns closely with customer-facing service intensity.

Service Evaluation

Chart Type: Stacked Bar Chart

This chart evaluates service experience challenges by category, segmented into delays, environmental discomfort, and overcrowding.

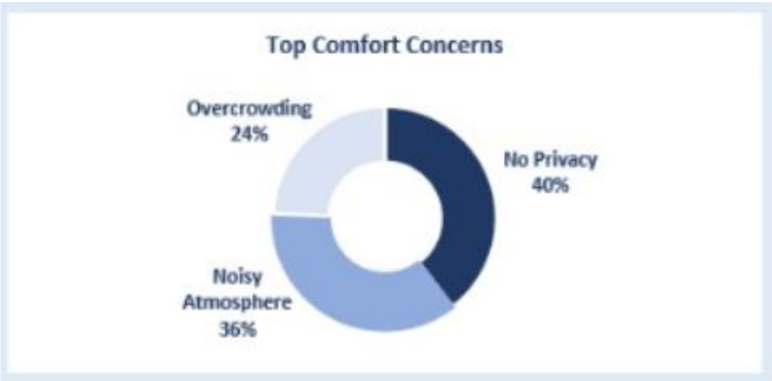
Beauty salons record high overcrowding (**20**) and notable delays (**12**).
Spa and wellness centers show significant environmental discomfort (**17**).

Medical clinics experience more delays (**14**) than comfort-related issues.
Physiotherapy and massage therapy studios show moderate but consistent service challenges.

Service experience issues vary significantly across environments.

Top Comfort Concerns

Chart Type: Donut Chart



This visualization highlights primary comfort-related concerns affecting customers.

No Privacy is the dominant concern at **40 percent**.
Noisy Atmosphere follows at **36 percent**.
Overcrowding contributes **24 percent**.

Environmental comfort strongly influences customer perception and satisfaction.

Booking Challenges Frequency

Chart Type: Frequency-Based Visualization

This visual categorizes booking challenges into high, moderate, and low frequency.

High-frequency challenges occupy the largest portion of the chart, indicating recurring scheduling issues.
Moderate-frequency challenges reflect operational inconsistencies.
Low-frequency challenges appear minimal in comparison.

Repeated booking failures point to systemic process gaps.

Loyalty Factors Driving Appointify Adoption

Chart Type: Area Chart

This chart shows how loyalty factors rated from low to high influence adoption intent.

Higher-rated loyalty factors dominate the chart area, showing stronger impact on adoption.
Lower-rated factors gradually decline, indicating weaker influence.

Adoption is driven by clearly perceived value rather than marginal benefits.

Pricing Perception

Chart Type: Grouped Bar Chart

This chart compares pricing-related concerns across service categories.

Unclear pricing is most pronounced in spa and wellness centers.
Beauty salons show notable levels of payment disputes.
Medical and physiotherapy services record moderate but persistent pricing concerns.

Pricing transparency emerges as a critical trust factor.

Top Booking Challenges

Chart Type: Vertical Bar Chart

This chart ranks booking challenges by frequency.

No Reminders leads with **114 responses**.
Mixed-Up Appointments follow at **112**.
Unclear Availability records **110**.
Poor Reachability accounts for **106**, while Long Waiting times record **101**.

Communication and scheduling gaps dominate customer pain points.

Appointify Readiness Scoreboard

Chart Type: Visual Scorecard

This scorecard summarizes customer readiness across service types.

Beauty Salon records a readiness score of **237**.
Spa and Wellness Center follows with **211**.
Massage Therapy Studio records **164**.

These segments represent the strongest early adopters.

Summary Interpretation

The Excel Customer Analysis dashboard shows that customers demonstrate high urgency and strong willingness to adopt Appointify, particularly within beauty and wellness services. However, adoption intent is constrained by manual booking practices, lack of reminders, pricing opacity, and environmental discomfort. The dashboard reinforces the need for structured scheduling, automated communication, transparent pricing, and customer-centric service design.

Power BI Dashboard Visualization

Appointify Business Analysis 2025 (Dashboard)

The Power BI dashboard presents an interactive analytical view that combines customer behaviour, operational readiness, booking efficiency, and technology adoption indicators. It enables stakeholders to assess how booking structures, reminder systems, feedback processes, and staffing coordination influence customer turnout, ratings, and willingness to adopt Appointify.

KPI Overview

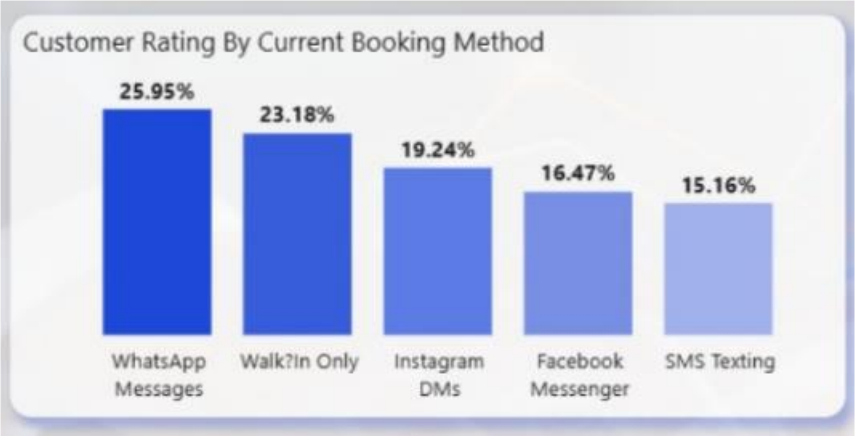
At the top of the dashboard, key performance indicators summarize the current state of appointment management from a customer-impact perspective.

High Turnout Percentage is recorded at **41.32 percent**, indicating that less than half of appointments result in strong attendance outcomes.
Low Turnout Percentage stands at **23.14 percent**, highlighting a significant portion of appointments with weak attendance.
No-Show Percentage is **1.65 percent**, reflecting a relatively small but still impactful level of missed appointments.
Technology Adoption Score averages **2.70**, suggesting moderate digital readiness among businesses interacting with customers.
Urgency Level Score is **2.47**, confirming that customer booking needs are generally time-sensitive.

These KPIs collectively indicate operational strain, moderate technology adoption, and high urgency expectations from customers.

Customer Rating by Current Booking Method

Chart Type: Vertical Bar Chart



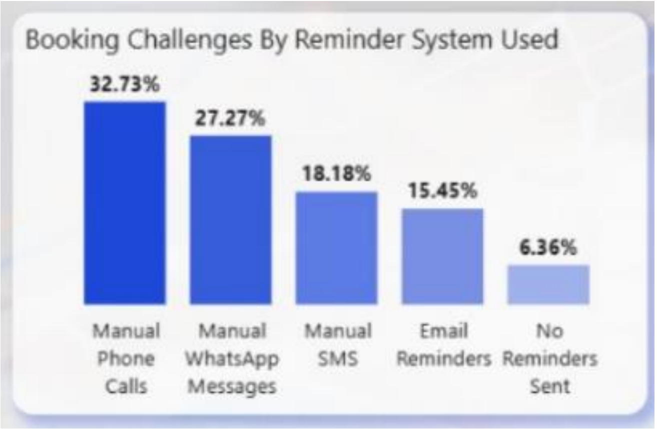
This chart evaluates customer satisfaction across different booking channels.

WhatsApp Messages receive the highest customer rating at **25.95 percent**, confirming it as the most trusted booking channel.
Walk-in Only bookings follow at **23.18 percent**, reflecting familiarity but limited structure.
Instagram DMs account for **19.24 percent**, while Facebook Messenger records **16.47 percent**.
SMS Texting ranks lowest at **15.16 percent**, suggesting limited engagement and effectiveness.

The chart shows that conversational and immediate channels perform better than fragmented or delayed communication methods.

Booking Challenges by Reminder System Used

Chart Type: Vertical Bar Chart



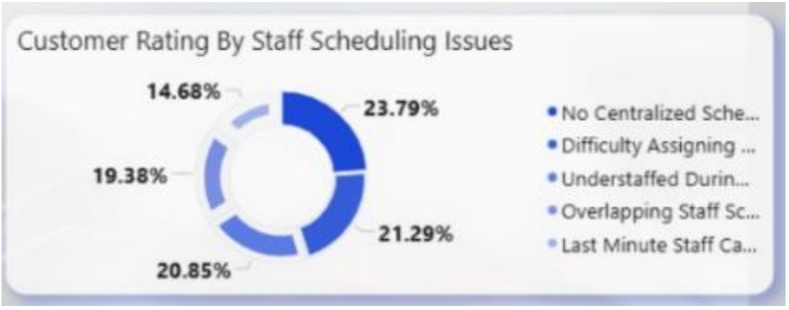
This visualization links booking challenges to the type of reminder system in use.

Manual Phone Calls account for the highest challenge rate at **32.73 percent**.
Manual WhatsApp Messages follow at **27.27 percent**.
Manual SMS contributes **18.18 percent**, while Email Reminders reduce challenges to **15.45 percent**.
Businesses that send no reminders experience the lowest representation at **6.36 percent**, though this reflects under-reporting rather than effectiveness.

The chart clearly demonstrates that automated reminders reduce booking challenges compared to manual approaches.

Customer Rating by Staff Scheduling Issues

Chart Type: Donut Chart



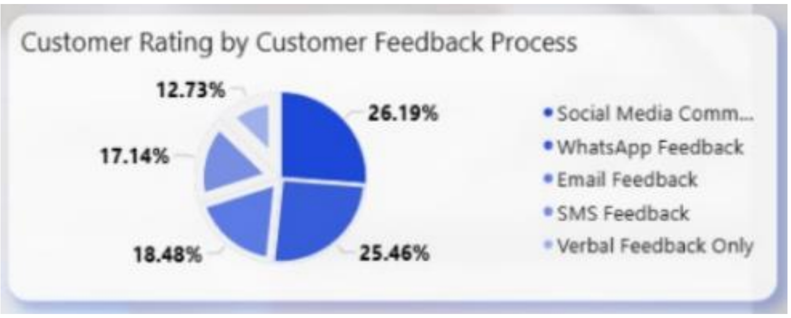
This chart shows how staff coordination problems affect customer ratings.

No centralized scheduling system represents **23.79 percent** of customer dissatisfaction.
Difficulty assigning staff accounts for **21.29 percent**.
Understaffing during peak periods contributes **20.85 percent**.
Overlapping staff schedules represent **19.38 percent**, while last-minute staff cancellations contribute **14.68 percent**.

Staff scheduling inefficiencies directly undermine customer experience and service reliability.

Customer Rating by Customer Feedback Process

Chart Type: Pie Chart



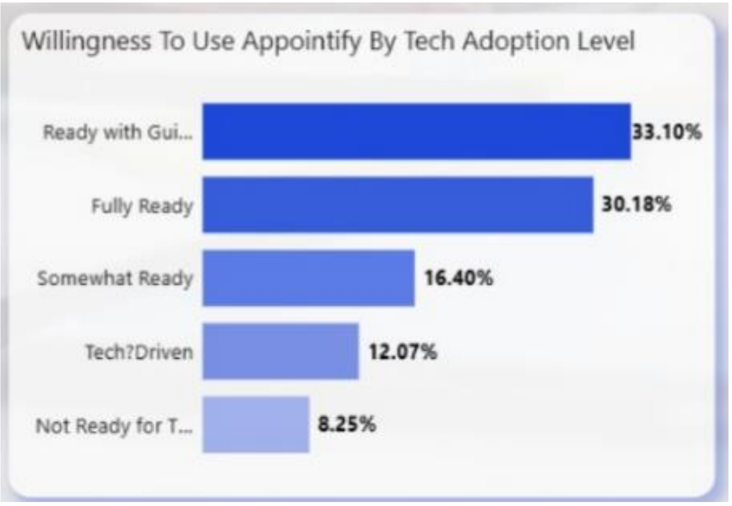
This chart evaluates how feedback collection methods influence customer ratings.

Social media communication leads with **26.19 percent**.
WhatsApp feedback follows closely at **25.46 percent**.
Email feedback contributes **18.48 percent**, while SMS feedback accounts for **17.14 percent**.
Verbal-only feedback ranks lowest at **12.73 percent**.

Digitally structured feedback channels correlate with higher customer satisfaction.

Willingness to Use Appointify by Tech Adoption Level

Chart Type: Horizontal Bar Chart



This visualization links digital readiness to adoption intent.

Businesses ready with guidance represent the highest willingness at **33.10 percent**.
Fully ready businesses follow at **30.18 percent**.
Somewhat ready businesses account for **16.40 percent**.
Tech-driven but fragmented users represent **12.07 percent**.
Not ready for technology adoption accounts for **8.25 percent**.

Willingness increases significantly as digital readiness improves.

Willingness to Use Appointify by Business Growth Goals

Chart Type: Table Visualization

Business Growth Goals	Willingness To Use Ap
Expand Service	**27.79%**
Increase Monthly Revenue	**23.40%**
Increase Customer Base	**17.02%**
Reduce Operational Costs	**16.09%**
Improve Customer Retention	**15.69%**

This table connects strategic goals to adoption intent.

Expanding services leads with **27.79 percent** willingness.
Increasing monthly revenue follows at **23.40 percent**.
Increasing customer base represents **17.02 percent**.
Reducing operational costs accounts for **16.09 percent**.
Improving customer retention records **15.69 percent**.

Growth-oriented goals strongly align with adoption interest.

Willingness to Use Appointify by Growth Rate

Chart Type: Table Visualization

Growth Rate	Willingness To Use App
Slow Growth	**32.39%**
Moderate Growth	**28.23%**
Rapid Growth	**15.75%**
Stagnant	**13.97%**
Declining	**9.66%**

This view compares adoption interest across growth stages.

Slow-growing businesses show the highest willingness at **32.39 percent**.
Moderate growth businesses follow at **28.23 percent**.
Rapid growth businesses represent **15.75 percent**.
Stagnant businesses account for **13.97 percent**, while declining businesses show the lowest interest at **9.66 percent**.

Businesses experiencing growth pressure are more motivated to adopt scheduling solutions.

Willingness to Use Appointify by Desired Features

Chart Type: Treemap

This chart highlights features that most influence adoption.

Customer history records lead at **25.05 percent**.
Payment tracking dashboard follows at **24.68 percent**.
No-show and cancellation control accounts for **17.79 percent**.
Staff schedule manager represents **16.88 percent**.
Online booking pages contribute **15.61 percent**.

Feature demand centers on visibility, control, and operational structure.

Dashboard Filters and Slicers

Interactive slicers on the left allow filtering by customer turnout, business type, booking method, booking challenges, reminder system, and willingness level. These enable dynamic exploration of relationships across operational and customer-focused variables.

Appointify Customer Analysis Dashboard, 2025 (Power BI)

This Power BI dashboard presents a focused analysis of customer behaviour, urgency levels, booking difficulties, and pricing concerns as they relate to the willingness to adopt Appointify. The dashboard emphasizes customer-side experience and readiness indicators.

KPI Overview

The KPI cards on the left summarize key customer indicators:

- **Willingness to Use Appointify: 3.28**
  This represents the average willingness score across all customer responses, indicating moderate-to-high readiness to adopt a centralized booking platform.

- **Service Type: 29**
  A total of 29 distinct service categories are represented in the customer dataset.

- **Environment & Comfort: 5**
  Five major comfort and environment-related concern categories were identified from customer responses.

Service Type Distribution

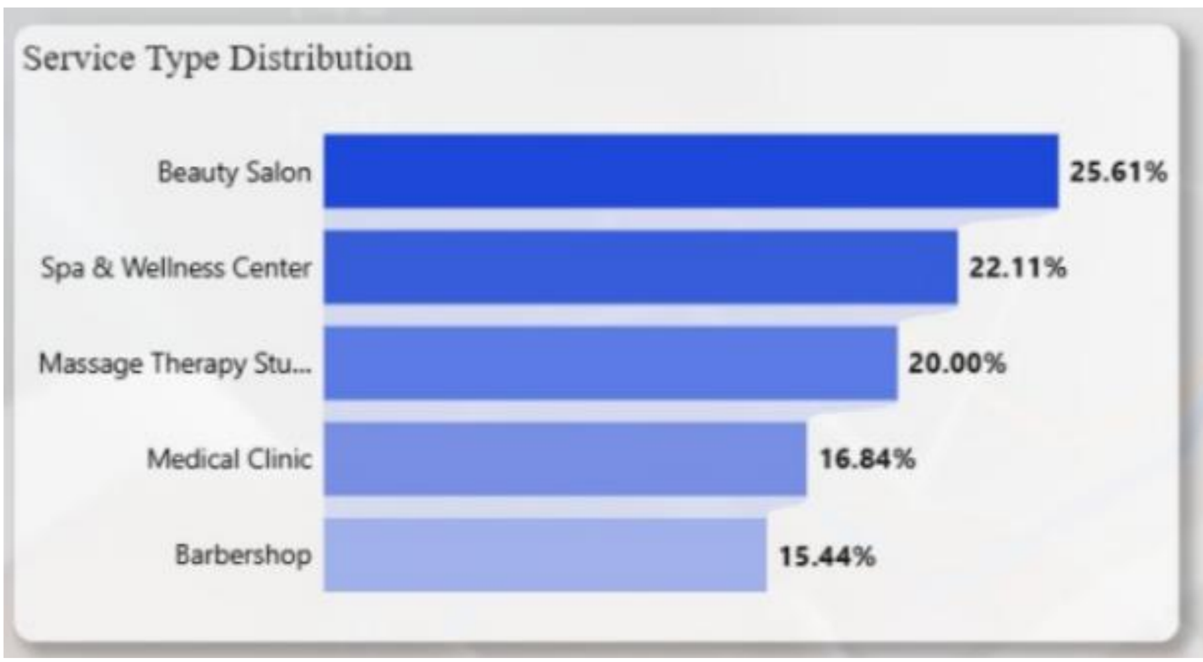


Chart Type: Horizontal Bar Chart

This chart shows the percentage distribution of customers across service types:

- **Beauty Salon: 25.61%**

- **Spa & Wellness Center: 22.11%**

- **Massage Therapy Studio: 20.00%**
- **Medical Clinic: 16.84%**
- **Barbershop: 15.44%**

Beauty salons represent the largest customer segment, followed closely by wellness-related services. These categories account for the majority of customer demand within the dataset.

Booking Difficulty by Willingness to Use Appointify

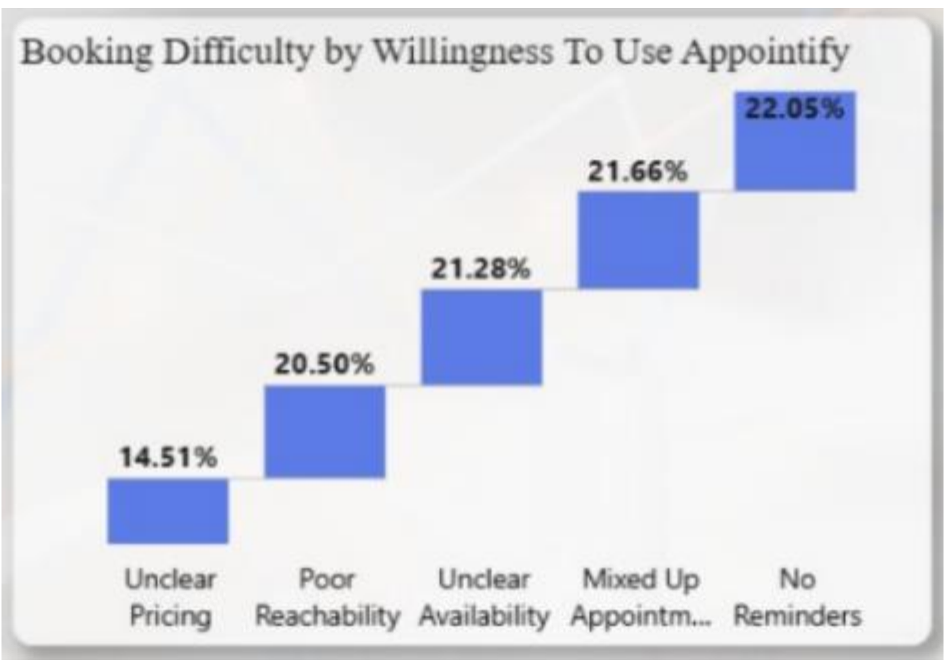


Chart Type: Waterfall Chart

This visualization shows how different booking challenges reduce customer willingness to use Appointify:

- **No Reminders: 22.05%**
- **Mixed-Up Appointments: 21.66%**
- **Unclear Availability: 21.28%**
- **Poor Reachability: 20.50%**
- **Unclear Pricing: 14.51%**

No reminders contribute the largest negative impact on willingness, confirming reminder absence as the most damaging booking issue from the customer perspective.

Payment / Pricing by Willingness to Use Appointify

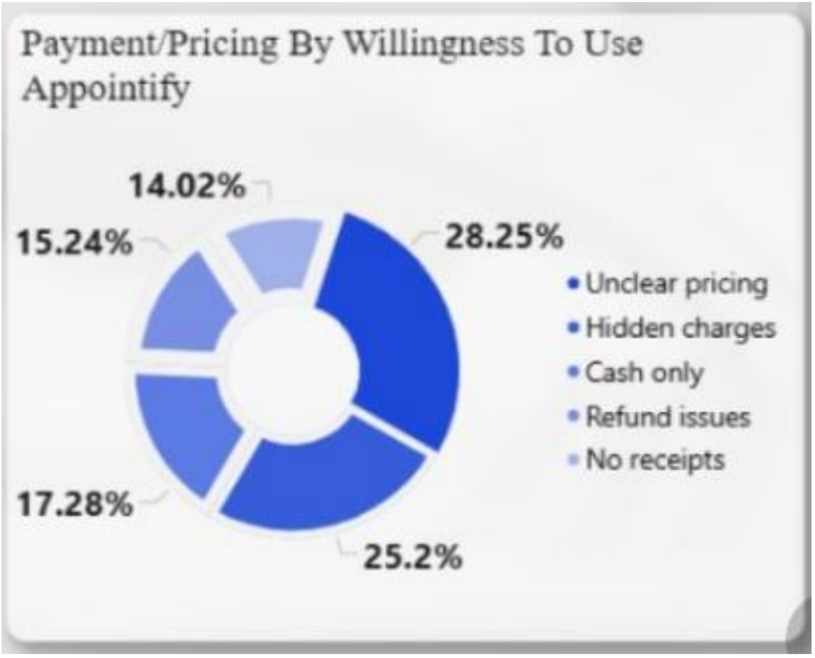


Chart Type: Donut Chart

This chart highlights pricing and payment issues affecting willingness:

- **Unclear Pricing: 28.25%**
- **Hidden Charges: 25.20%**
- **Cash Only: 17.28%**
- **Refund Issues: 15.24%**
- **No Receipts: 14.02%**

Pricing transparency issues collectively dominate customer concerns, with unclear pricing emerging as the most significant factor.

Urgency Level of Need by Willingness to Use Appointify

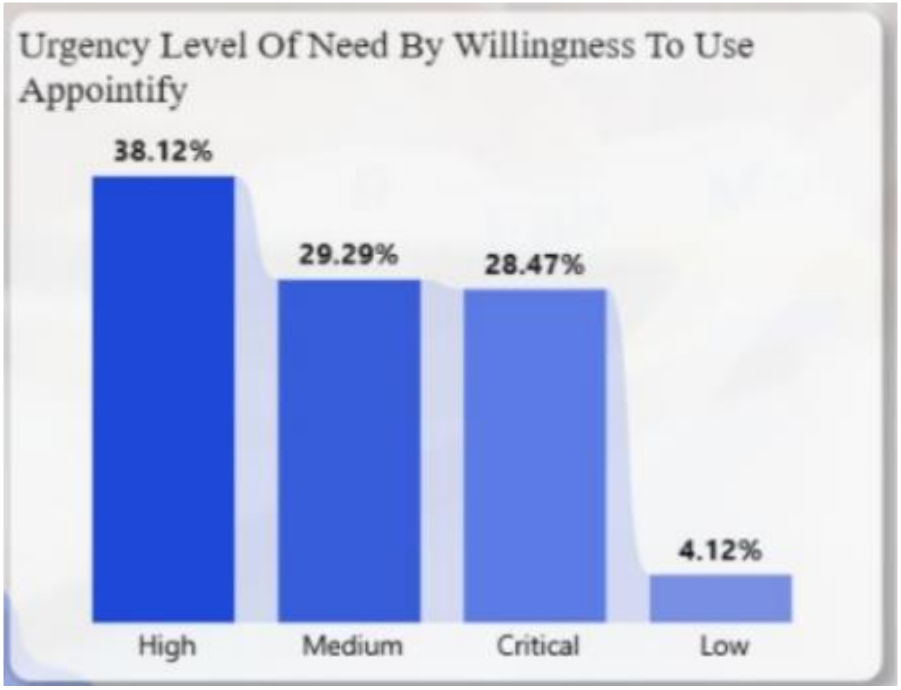


Chart Type: Vertical Bar Chart

This chart shows urgency levels among customers willing to use Appointify:

- **High Urgency: 38.12%**
- **Medium Urgency: 29.29%**
- **Critical Urgency: 28.47%**
- **Low Urgency: 4.12%**

Most customers fall within high or critical urgency categories, indicating strong demand for fast and reliable booking confirmations.

Urgency Level of Need by Access Level

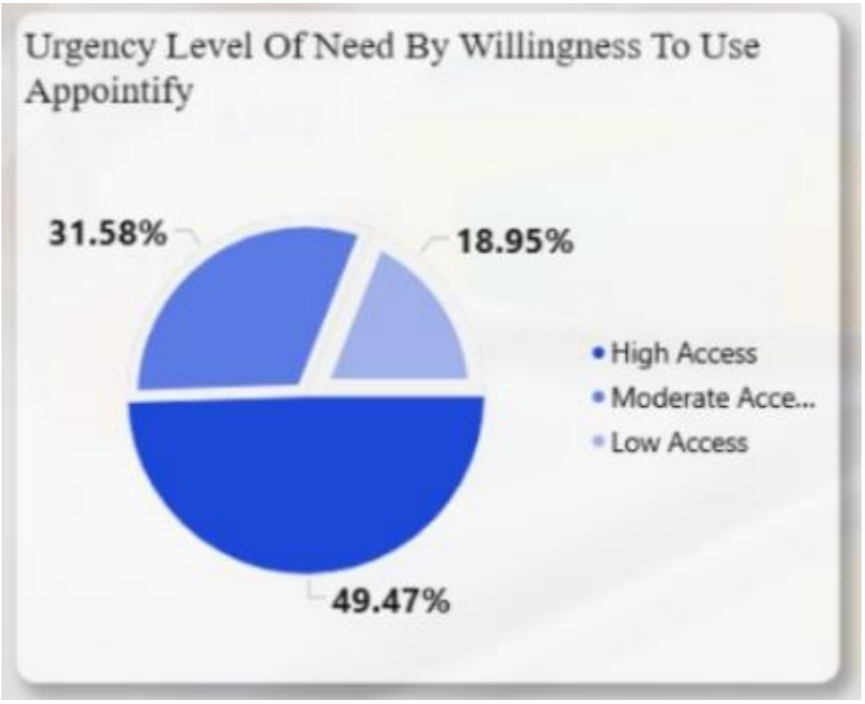


Chart Type: Pie Chart

This chart links urgency to access level:

- **High Access: 49.47%**
- **Moderate Access: 31.58%**
- **Low Access: 18.95%**

Nearly half of customers report high access, suggesting that availability exists but is not always efficiently managed.

Urgency Level Distribution

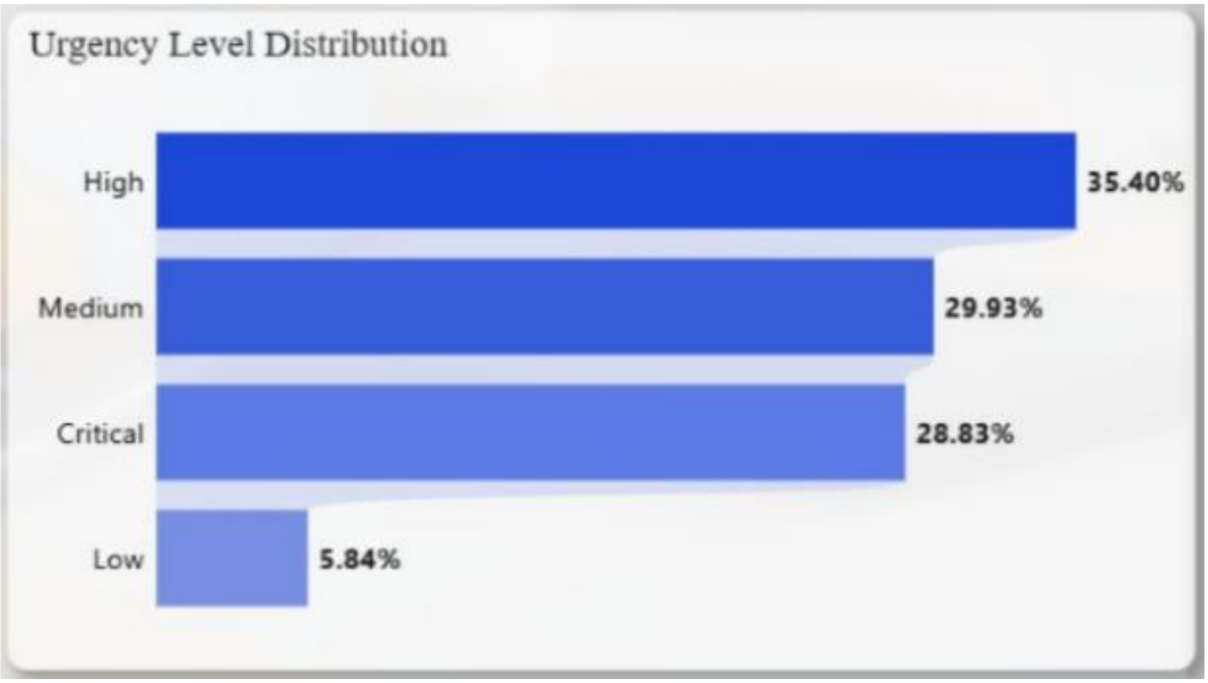


Chart Type: Horizontal Bar Chart

This chart provides an overall urgency distribution:

- **High: 35.40%**
- **Medium: 29.93%**
- **Critical: 28.83%**
- **Low: 5.84%**

High and critical urgency levels dominate, reinforcing the time-sensitive nature of customer booking behaviour.

Summary

The Appointify Customer Dashboard, 2025 clearly shows that customers exhibit **high urgency and moderate-to-high willingness** to adopt a centralized booking solution. However, this willingness is significantly reduced by **operational weaknesses**, particularly the absence of reminders, mixed-up appointments, unclear availability, and poor pricing transparency.

Beauty salons and wellness-related services dominate customer demand, making them ideal early adopters. Pricing clarity and reminder automation emerge as the strongest levers for improving customer confidence, satisfaction, and booking completion rates.

Overall, the dashboard demonstrates that **customer readiness exists**, but adoption is constrained by fragmented booking processes and inconsistent communication—gaps that Appointify is positioned to resolve.

9. Observations and Recommendations

Observations from the Business Dataset

The business dataset reveals a strong dependence on manual booking channels such as WhatsApp, phone calls, and walk-ins. These methods dominate appointment handling across most service types and are directly associated with higher booking errors, delayed confirmations, and operational stress.

Missed, late, and rescheduled appointments occur frequently. The booking outcome distribution shows that completed appointments represent a smaller proportion compared to rescheduled, late, and missed bookings. This indicates weak follow-through mechanisms and insufficient confirmation processes.

Slow response time consistently appears as the most frequent booking challenge. Businesses struggle to respond promptly to booking requests, particularly during peak operating hours, leading to customer dissatisfaction and drop-offs.

Double booking and overbooking are common outcomes of poor availability tracking. The absence of real-time scheduling and staff synchronization leads to conflicting appointments and service delays.

Customer information is poorly structured across most businesses. The strong demand for customer history records indicates that businesses currently lack centralized customer profiles, limiting personalization and service continuity.

Peak operating periods expose operational weaknesses. The peak hours analysis shows that difficulty levels rise sharply as booking volume increases, with manual systems failing under high demand.

Payment tracking is loosely connected to bookings. Businesses face manual entry errors, delayed confirmations, and rising disputes as weekly administrative hours increase, indicating inefficiency rather than improved control.

Business growth intent is high, particularly among beauty salons and wellness centers. However, growth is constrained by operational limitations rather than lack of demand or interest.

Technology adoption is driven by the need for efficiency rather than advanced features. Businesses show higher willingness to adopt tools that simplify operations and reduce workload.

Observations from the Customer Dataset

Customer willingness to use Appointify is high across most service categories, with beauty salons, spa and wellness centers, and massage therapy studios leading adoption readiness.

Urgency levels are predominantly high and critical. Customers expect fast confirmations and immediate booking clarity, indicating that delays directly undermine satisfaction.

No reminders emerge as the most significant customer-facing booking challenge. This aligns with high rates of missed and rescheduled appointments observed in the business dataset.

Comfort-related concerns strongly influence customer perception. Lack of privacy, noisy environments, and overcrowding collectively account for the majority of comfort complaints.

Pricing transparency remains a major issue. Unclear pricing, hidden charges, and cash-only systems reduce trust and negatively affect willingness to complete bookings.

Customers using more structured booking methods report higher satisfaction levels. Informal booking channels correlate with lower ratings due to slower responses and unclear confirmations.

There is a visible gap between booking intent and attendance. High willingness does not consistently translate into completed appointments, reinforcing the impact of poor booking structure.

Recommendations Based on Business and Customer Insights

A centralized digital booking system should be introduced to consolidate all appointments into a single real-time schedule. This will eliminate fragmented booking records and reduce conflicts.

Automated appointment confirmations and reminders should be implemented using SMS, WhatsApp, and email. This directly addresses the leading causes of no-shows and missed appointments.

Real-time availability and staff scheduling synchronization should be enabled to prevent double booking and overbooking. Staff capacity should be visible at the point of booking.

Structured customer profiles should be provided, capturing appointment history, preferences, and contact details. This supports personalization, trust, and repeat visits.

Peak-hour booking controls should be introduced, including booking caps, buffer times, and capacity alerts. This reduces chaos during high-demand periods.

Online booking links should be made easily shareable across WhatsApp and social platforms to reduce reliance on manual conversations.

Payment tracking should be fully integrated with appointment records. Payment status, confirmations, and disputes should be linked directly to bookings.

Clear cancellation and rescheduling rules should be enforced to reduce last-minute changes and improve service predictability.

Basic analytics dashboards should be provided to business owners, showing no-show rates, cancellations, repeat customers, and service performance.

The system should be designed with a mobile-first approach, allowing business owners to manage bookings conveniently on the go.

Onboarding and interface design should prioritize simplicity, minimizing learning effort for non-technical users.

Appointify should be positioned as an operational assistant rather than a complex software tool, emphasizing time savings, organization, and reliability.

Final Recommendations

Based on the combined insights from the business and customer datasets, the analysis reveals clear operational gaps, behavioral patterns, and adoption opportunities within appointment-based service businesses. The following recommendations are designed to address these findings holistically, aligning customer expectations with business operational needs while positioning Appointify as a scalable, value-driven solution.

Recommendations for Appointment-Based Businesses

Service-based businesses, particularly beauty salons, spa and wellness centers, massage therapy studios, and similar high-frequency service providers, should transition from fragmented, manual booking processes to a centralized digital appointment management system. The heavy reliance on WhatsApp messages, phone calls, and walk-ins has been shown to increase scheduling conflicts, slow response times, and customer dissatisfaction. Consolidating all booking channels into a single, real-time scheduling interface will significantly reduce double bookings, misplaced appointments, and response delays.

Automated appointment confirmations and reminders should be adopted as a standard operational practice. The high occurrence of no-shows and rescheduled appointments across both datasets indicates that manual follow-ups are insufficient and inconsistent. Businesses should implement automated reminders via WhatsApp, SMS, or email at predefined intervals before appointments to improve attendance rates and reduce revenue loss.

Businesses should integrate staff scheduling directly with customer bookings. The analysis shows that staff availability is often disconnected from appointment intake, leading to idle staff periods or shortages during peak hours. Aligning bookings with staff capacity in real time will improve service flow, reduce customer waiting times, and optimize workforce utilization.

Payment tracking should be embedded within the booking workflow rather than managed as a separate process. The dashboards reveal frequent payment disputes, delayed confirmations, and manual entry errors, particularly among businesses spending extended hours managing payments. Linking payment status directly to appointment records will improve transparency, reduce disputes, and provide clearer financial oversight.

During peak operating periods, businesses should implement booking limits, buffer times, and queue controls. The peak hours analysis highlights increased chaos, missed bookings, and operational strain when demand is high. Introducing controlled slot allocation and automated load balancing during these periods will help maintain service quality and reduce customer frustration.

Customer data management should be prioritized through structured customer profiles. Businesses lacking customer history records struggle with personalization, follow-ups, and retention. Maintaining appointment history, preferences, and contact information will enhance service consistency and build customer trust.

Recommendations by Business Segment

Beauty salons and spa and wellness centers, which show the highest growth rates, customer turnout, urgency levels, and willingness to adopt Appointify, should be prioritized for early adoption and full feature deployment. These segments benefit most from automated scheduling, reminders, payment tracking, and customer history tools due to their high booking volumes and repeat customer base.

Massage therapy studios and similar mid-volume service providers should focus on improving reminder systems and availability management to reduce rescheduling and late arrivals, which are more prevalent in these categories.

Medical clinics and physiotherapy centers, which show more cautious adoption behavior, should be introduced to simplified booking workflows that complement existing structured systems. Emphasis should be placed on appointment adherence, urgency tagging, and payment transparency rather than volume-driven features.

Recommendations Based on Customer Insights

Customers place high value on speed, clarity, and reliability in the booking process. The dominance of high and critical urgency requests indicates that customers often require immediate confirmation and fast response. Businesses should ensure that booking requests are acknowledged instantly, even if confirmation follows later.

Comfort-related concerns, particularly lack of privacy, overcrowding, and noisy environments, should be addressed through better appointment spacing and reduced walk-in dependency. Structured scheduling can help control crowd density and improve the overall service environment.

Pricing transparency must be improved across service categories. Unclear pricing and hidden charges are leading contributors to customer dissatisfaction. Businesses should provide clear pricing information at the point of booking and link payment receipts directly to appointments.

Structured booking methods consistently receive higher customer ratings than informal messaging channels. Businesses should encourage customers to use official booking links or platforms rather than relying solely on direct messages, which are prone to delays and miscommunication.

Product-Level Recommendations for Appointify

Appointify should position itself as an operational assistant rather than just a booking tool. The platform should emphasize time savings, organization, reliability, and reduced stress for business owners, as these factors strongly influence adoption decisions.

Feature development should prioritize automated reminders, real-time availability locking, customer profile management, and integrated payment tracking, as these directly address the most frequent operational and customer pain points identified in the analysis.

Onboarding and user experience should remain simple and intuitive. The data indicates that digital adoption among small businesses is driven by efficiency rather than feature complexity. A low learning curve will accelerate adoption, particularly among non-technical users.

Analytics provided to business owners should focus on actionable performance indicators such as no-show rates, cancellation trends, repeat customers, and peak-hour performance. These insights will enable data-driven decision-making and reinforce the value of the platform.

Alignment Between Business and Customer Needs

The comparative analysis of both datasets shows strong alignment between customer expectations and business challenges. Customers demand faster responses, clearer communication, and reliable scheduling, while businesses struggle with manual coordination, missed appointments, and operational overload. Appointify directly bridges this gap by offering a unified system that improves efficiency, enhances customer experience, and supports sustainable business growth.

By addressing both sides of the appointment lifecycle, from booking intent to service completion, the recommended strategies position Appointify as a scalable solution capable of transforming appointment management within the service industry.

10. Comparative Analysis of Business and Customer Datasets

The business and customer datasets complement each other by presenting both operational inputs and experiential outcomes within appointment-based services.

From the business perspective, operational inefficiencies such as manual booking, poor reminder systems, and weak payment tracking are clearly documented. These inefficiencies manifest as slow response times, booking conflicts, and high administrative workload.

From the customer perspective, the consequences of these inefficiencies are visible through high urgency requests, frequent complaints about no reminders, unclear pricing, and lower satisfaction levels.

Both datasets consistently point to the same friction points. Manual processes increase workload for businesses while simultaneously reducing customer trust and attendance. The high willingness to adopt Appointify observed in both datasets indicates that the problem is not resistance to technology, but the lack of a suitable, simple solution.

The business dataset highlights the internal struggle to scale operations efficiently, while the customer dataset highlights unmet expectations around speed, clarity, and reliability. Together, they validate the need for a centralized scheduling platform that aligns business operations with customer expectations.

The convergence of high customer urgency and high business workload reinforces the core value proposition of Appointify. By addressing operational gaps identified in the business dataset, the platform directly improves customer outcomes observed in the customer dataset.

Overall, the combined analysis confirms that improved booking structure, automation, and visibility benefit both sides of the service relationship, supporting sustainable growth, higher satisfaction, and operational efficiency.

11. Project Summary

This capstone project examined appointment management practices within small, service-based businesses through the development and analysis of the Appointify digital scheduling concept. Using both business and customer datasets, the project explored how manual booking processes, communication gaps, and limited use of automation affect operational efficiency, customer experience, and business growth.

The analysis combined structured survey data, Excel-based data cleaning and analysis, and dashboard-driven insights developed using Microsoft Excel and Power BI. Key areas of focus included booking challenges, reminder systems, payment tracking, customer turnout, urgency levels, and willingness to adopt a centralized appointment system. The findings consistently highlighted inefficiencies in manual workflows and demonstrated strong readiness for a digital scheduling solution, particularly among beauty, wellness, and lifestyle-based service providers.

12. Conclusion

Key Learnings

The analysis revealed that manual appointment management remains the dominant practice across small service-based businesses, relying heavily on WhatsApp messages, phone calls, and walk-ins. While these methods are familiar, they contribute significantly to operational strain, booking errors, missed appointments, and inconsistent customer experiences.

Customer data showed high urgency in booking needs and a strong willingness to use a digital solution, especially in high-frequency service categories such as beauty salons, spa and wellness centers, and massage therapy studios. However, this willingness does not consistently translate into completed appointments due to the absence of automated reminders, unclear availability, and slow response times.

From the business perspective, growth ambitions are clearly centered on service expansion and revenue increase, yet operational inefficiencies such as poor payment tracking, fragmented customer records, and unsynchronized staff scheduling limit scalability. Across both datasets, structured booking processes were strongly associated with higher customer turnout, better ratings, and reduced booking challenges.

Collectively, the findings validate the need for a centralized, automated appointment management system like Appointify to bridge the gap between customer intent and successful service delivery.

13. Limitations

Despite providing valuable insights, the analysis is subject to certain limitations. The datasets were derived from survey responses, which introduces self-reporting bias and relies on participants' understanding of operational and technological concepts. The sampling approach, while broad, was influenced by network-based distribution, which may limit full representation of all service industry segments.

Additionally, the analysis focused on cross-sectional data collected over a limited time frame, restricting the ability to observe long-term behavioral trends or seasonal effects. Advanced statistical modeling was outside the scope of this project, as the primary emphasis was on descriptive and comparative analysis using spreadsheet and dashboard tools.

Future Research

Future work could expand this analysis by incorporating longitudinal data to track changes in booking behavior, customer retention, and revenue performance over time. Integrating transactional data, such as actual booking logs and payment records, would allow for deeper performance measurement and predictive modeling.

Further research could also explore geographic segmentation, pricing sensitivity analysis, and feature adoption patterns after implementation. From a product perspective, evaluating post-deployment usage data would provide insights into feature effectiveness, system usability, and long-term business impact.

14. References

The project analysis was supported by the following tools and data sources:

Microsoft Excel for data cleaning, preprocessing, descriptive analysis, pivot tables, and dashboard creation
Microsoft Power BI for interactive customer dashboard visualization and KPI tracking
Primary survey data collected for the Appointify Business Plan and Customer Plan datasets

Internal project documentation and structured survey instruments designed for this capstone project

15. Appendices

The appendices include supporting materials referenced throughout the analysis, such as:

Detailed Excel pivot tables used for business and customer analysis
Dashboard screenshots for business and customer insights
Additional charts and visual breakdowns not included in the main report body
Supplementary explanations of data preparation steps and Excel functions applied during analysis

These materials provide additional context and transparency for readers seeking deeper technical understanding of the analytical process.